

Exp 4
Date 25/7/26

Frequency Modulation (FM) signal Generation And Bandwidth Analysis

AIM:

To generate and Analyze a frequency Modulated (FM) signal using Scilab by observing the message signal, carrier signal, and the FM waveform.

APPARATUS REQUIRED:

- * Computer / Laptop
- * Scilab software

CODE :

```
clc;  
clear;  
close;  
am = 4;  
fm = 200;  
ac = 18;  
fc = 2000;  
B = 5;  
fs = 20000;
```

SNO	Particulars	Theory		Practical	
		Amplitude	RF frequency	Amplitude	Time $T = t_2 - t_1$
1	Message signal	4	200	4	0.0049 S
2	Carrier Signal	8	2000	8	$T = t_2 - t_1$ 0.000491 S
3	FM Signal	-		8.9 - 8.9	$T_{max} = 0.000$ 490 $t_{min} = 0.000$ 490
					2040.8 Hz 3056.8 Hz 1008 Hz

$\Delta f = k_f A_m$
 $\Delta f = \beta f_m$
 $= 5 \times 200 = 1000 \text{ Hz}$
 $f_{max} = f_c + \Delta f = 2000 + 1000$
 $= 3000 \text{ Hz}$
 $f_{min} = f_c - \Delta f$
 $= 2000 - 1000$
 $= 1000 \text{ Hz}$

$f_{min} = \frac{1}{T_{max}}$
 $f_{max} = \frac{1}{T_{min}}$

```
t = 0:1/fs:2/fm;
```

```
em = am * sin(2 * %pi * fm * t);
```

```
ec = ac * sin(2 * %pi * fc * t);
```

```
efm = ac * cos((2 * %pi * fc * t) + B * sin(2 * %pi * fm * t));
```

```
figure(1);
```

```
subplot(3,1,1);
```

```
plot(t, em);
```

```
xgrid();
```

```
title("Message signal");
```

```
xlabel("Time (s)");
```

```
ylabel("Amplitude");
```

```
subplot(3,1,2);
```

```
plot(t, ec);
```

```
xgrid();
```

```
title("carrier signal");
```

```
xlabel("Time (s)");
```

```
ylabel("Amplitude");
```

```
subplot(3,1,3);
```

```
plot(t, efm);
```

```
xgrid();
```

```
title("Frequency modulated (Fm) signal");
```

```
xlabel("time (s)");
```

```
ylabel("Amplitude");
```

Calculation:

Message signal:

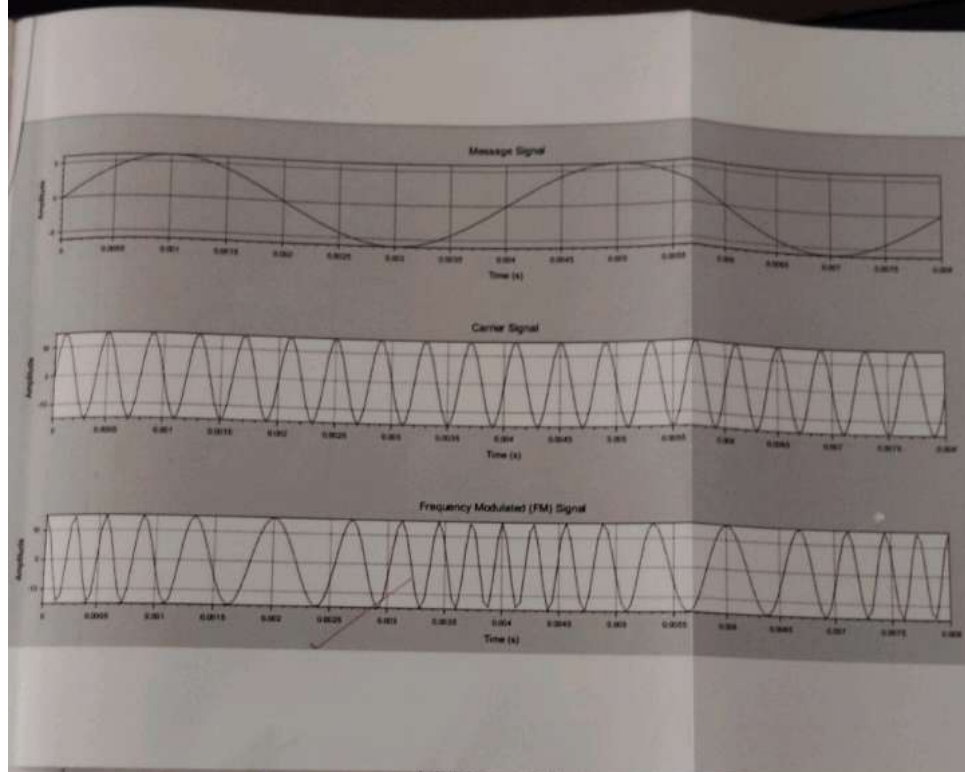
$$\begin{aligned}T &= x_2 - x_1 \\&= (5.91 \times 10^{-3}) - (1.00 \times 10^{-3}) \\&= 4.9 \times 10^{-3} \\&= 0.0049\end{aligned}$$

$$\begin{aligned}f &= 1/T \\&= 1/0.0049 \\&= 204.1 \text{ Hz}\end{aligned}$$

Carrier signal

$$\begin{aligned}T &= x_2 - x_1 \\&= (6.9 \times 10^{-4}) - (2.00 \times 10^{-4}) \\&= 4.9 \times 10^{-4} \\&= 0.00049\end{aligned}$$

$$\begin{aligned}f &= 1/T \\&= 1/4.9 \times 10^{-4} \\&= 2040.8 \text{ Hz}\end{aligned}$$



Thus, the frequency modulation (FM) signal was successfully generated using Matlab. The message signal, carrier signal and FM signal were observed.

Marks allocation:

Scilab program implementation (20)	wave form and tabulation (20)	Calculation and result (20)	viva - voice (20)	Record observation (20)	Total marks.
20	20	20	17	20	97

RESULT:

Thus, the frequency modulation (FM) signal was successfully generated using Scilab. The message signal, carrier signal and FM signal were observed.



FM signal.

Case 1.

$$\begin{aligned}\Delta f &= f_c - f_{\min} \\ &= 2000 - 1008.1 \\ &= 991.9\end{aligned}$$

$$\begin{aligned}\beta &= \frac{\Delta f}{f_m} \\ &= \frac{991.9}{200} \\ &= 4.9595\end{aligned}$$

Case 2.

$$\begin{aligned}\Delta f &= f_{\max} - f_c \\ &= 3056.2 - 2000 \\ &= 1056\end{aligned}$$

$$\begin{aligned}\beta &= \frac{\Delta f}{f_m} \\ &= \frac{1056}{200} \\ &= 5.28\end{aligned}$$

$$T_{\min} = \frac{1}{f_{\max}} = \frac{1}{2040.8} = 0.000490$$

$$T_{\max} = \frac{1}{f_{\min}} = \frac{1}{2041} = 0.0004901$$